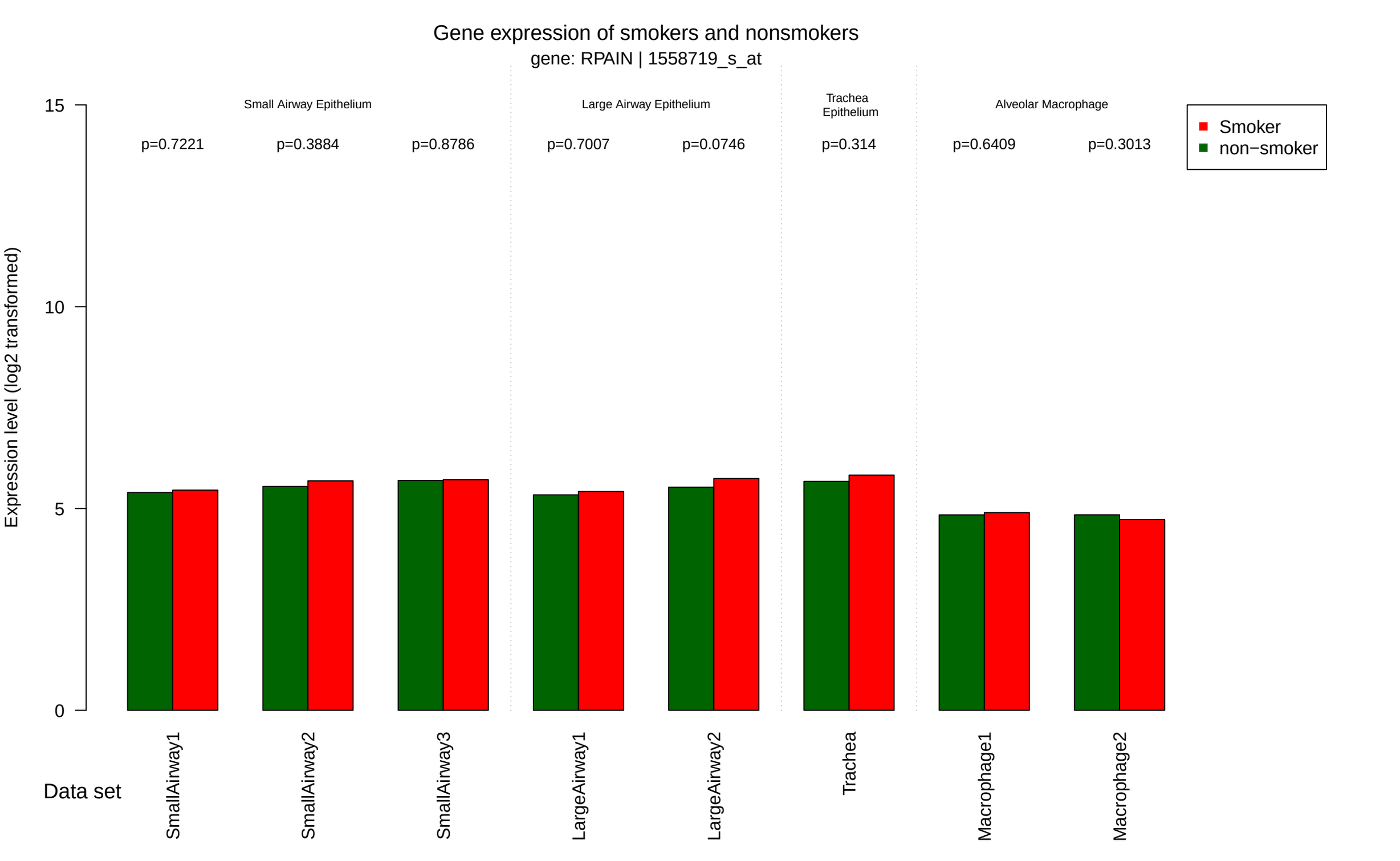
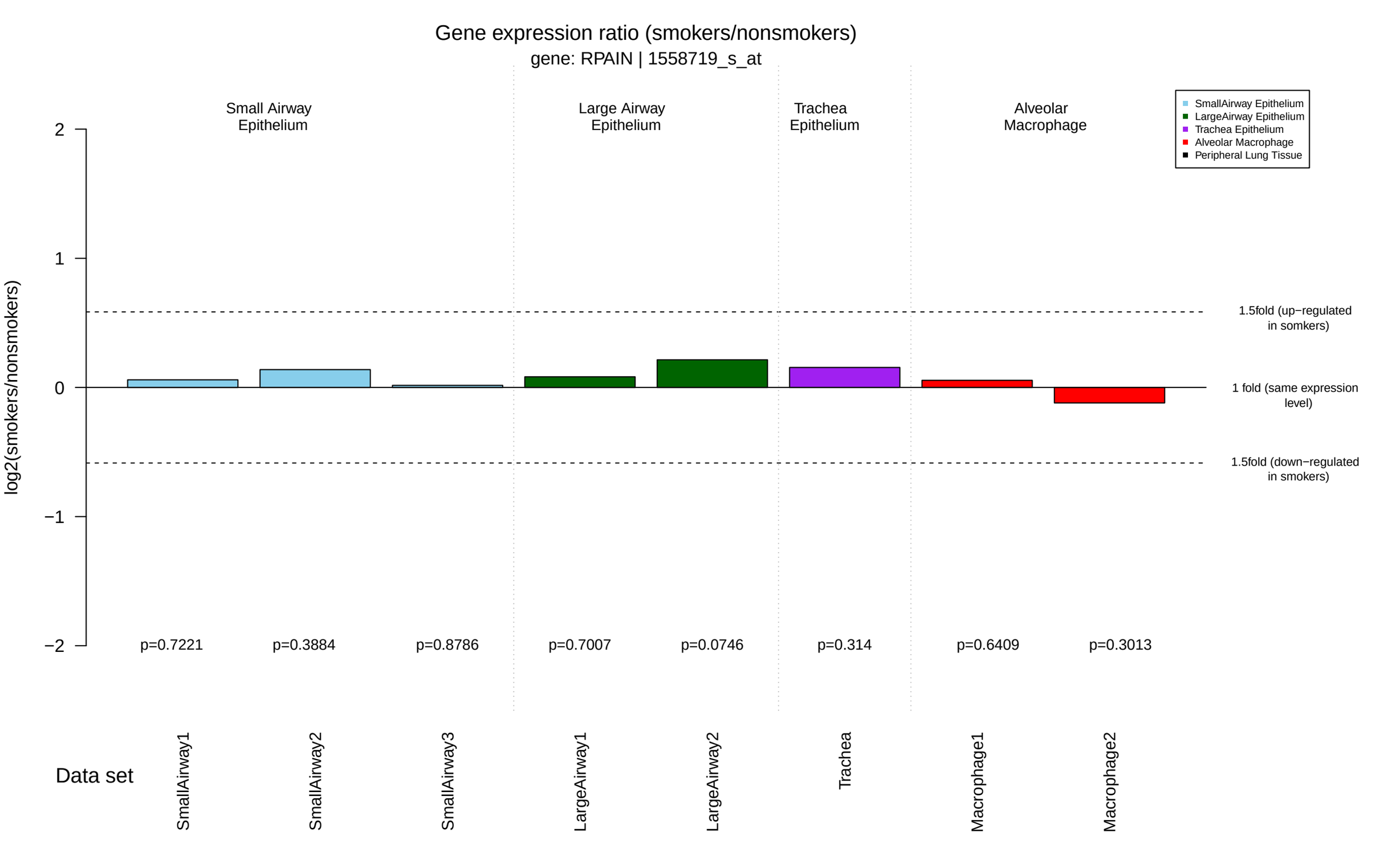
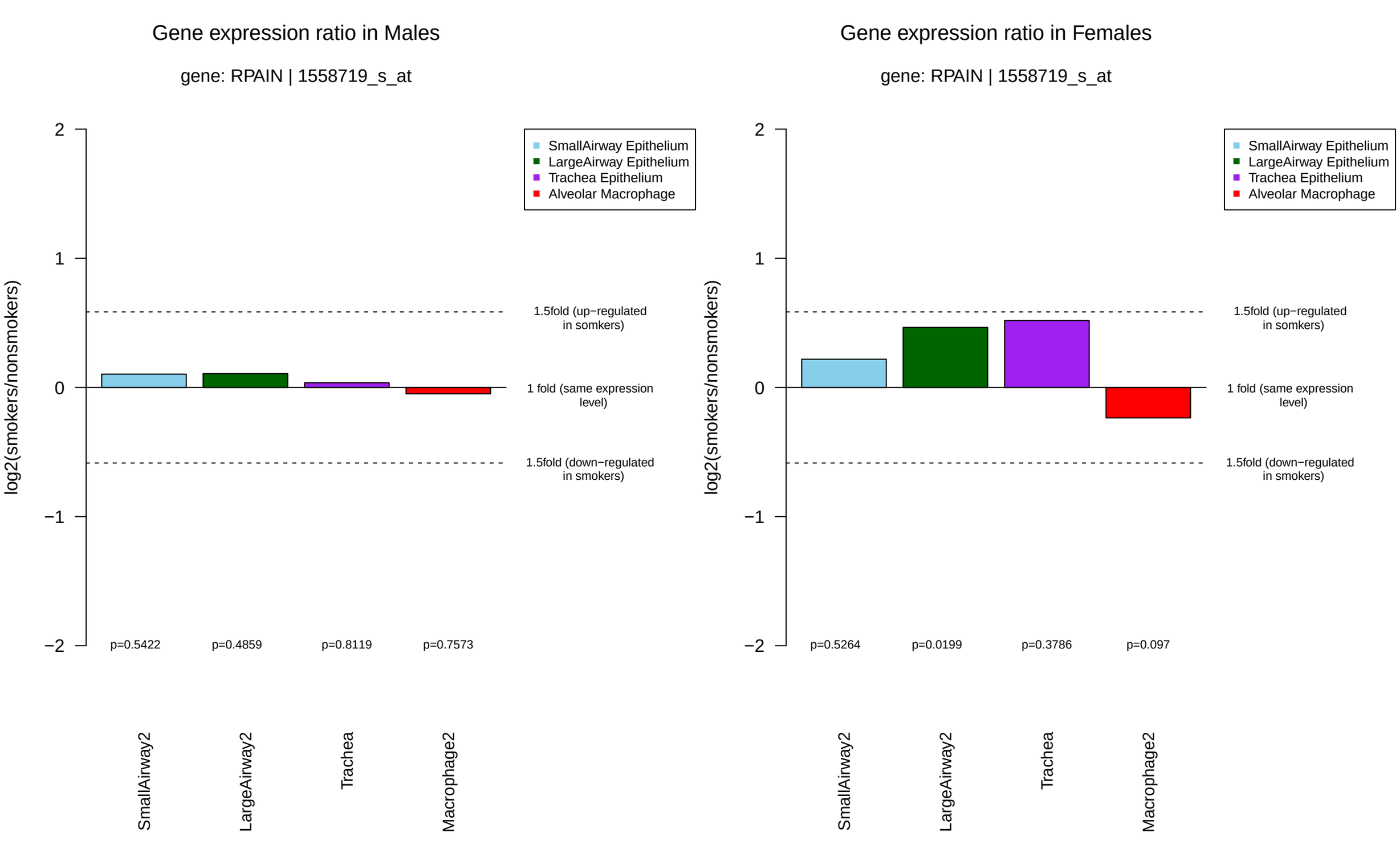
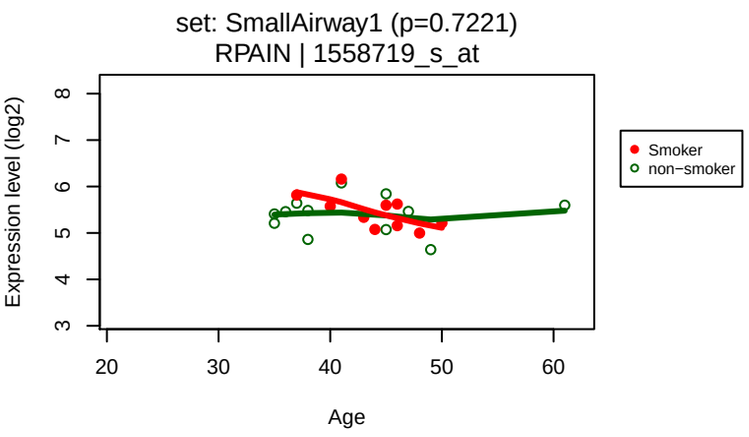
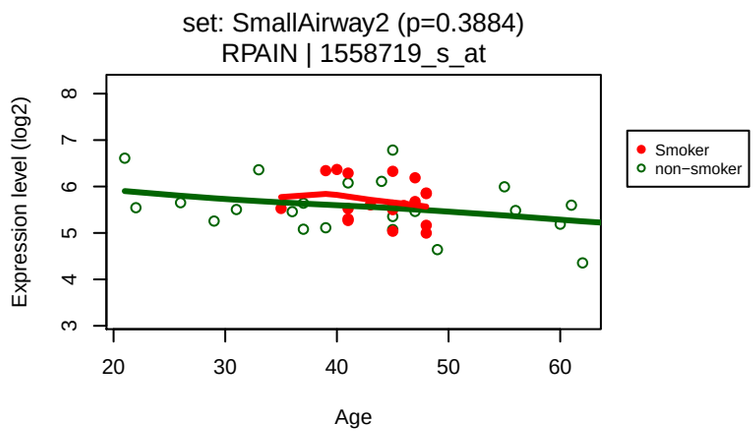
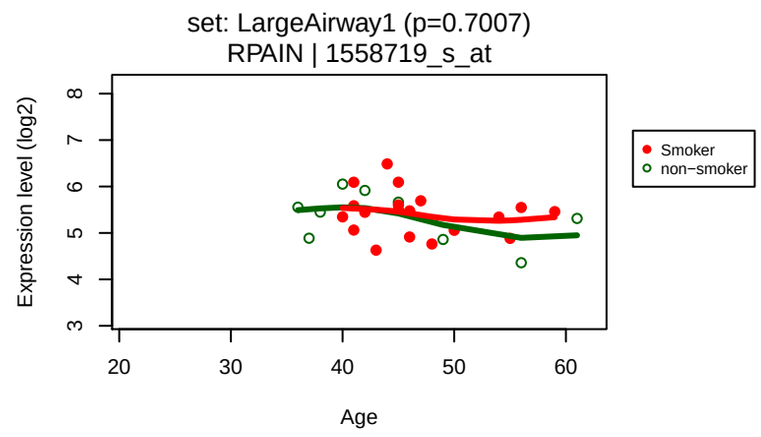

Gene expression of smokers and nonsmokers
gene: RPAIN | 1558719_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.7221
p=0.3884
p=0.8786
p=0.7007
p=0.0746
p=0.314
p=0.6409
p=0.3013
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

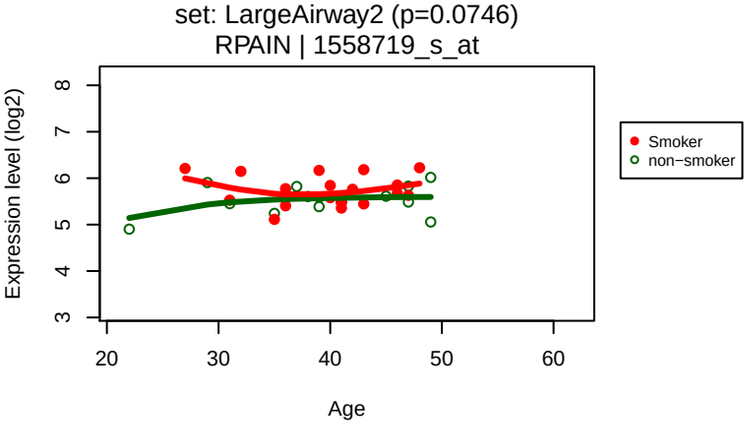

Gene expression ratio (smokers/nonsmokers)
gene: RPAIN | 1558719_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.7221
p=0.3884
p=0.8786
p=0.7007
p=0.0746
p=0.314
p=0.6409
p=0.3013
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

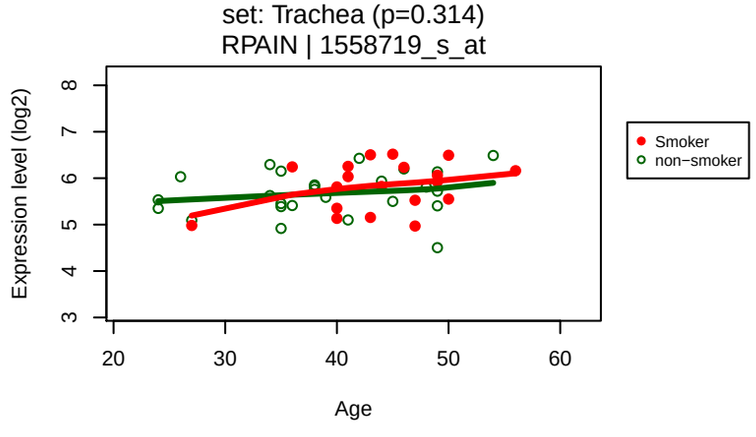

Gene expression ratio in Males
gene: RPAIN | 1558719_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5422
p=0.4859
p=0.8119
p=0.7573
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: RPAIN | 1558719_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5264
p=0.0199
p=0.3786
p=0.097
SmallAirway2
LargeAirway2
Trachea
Macrophage2

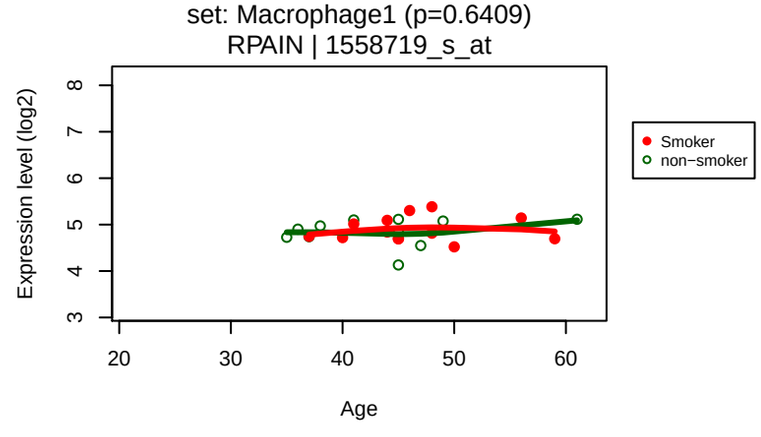

set: SmallAirway1 (p=0.7221)
RPAIN | 1558719_s_at
Expression level (log2)
Age
Smoker
non−smoker

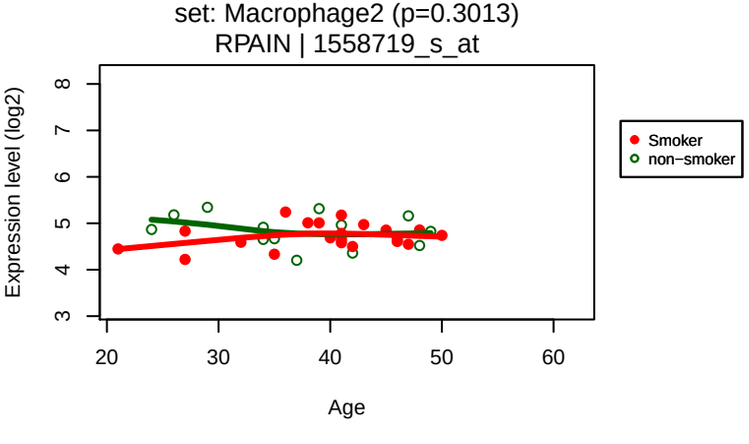

set: SmallAirway2 (p=0.3884)
RPAIN | 1558719_s_at
Expression level (log2)
Age
Smoker
non−smoker

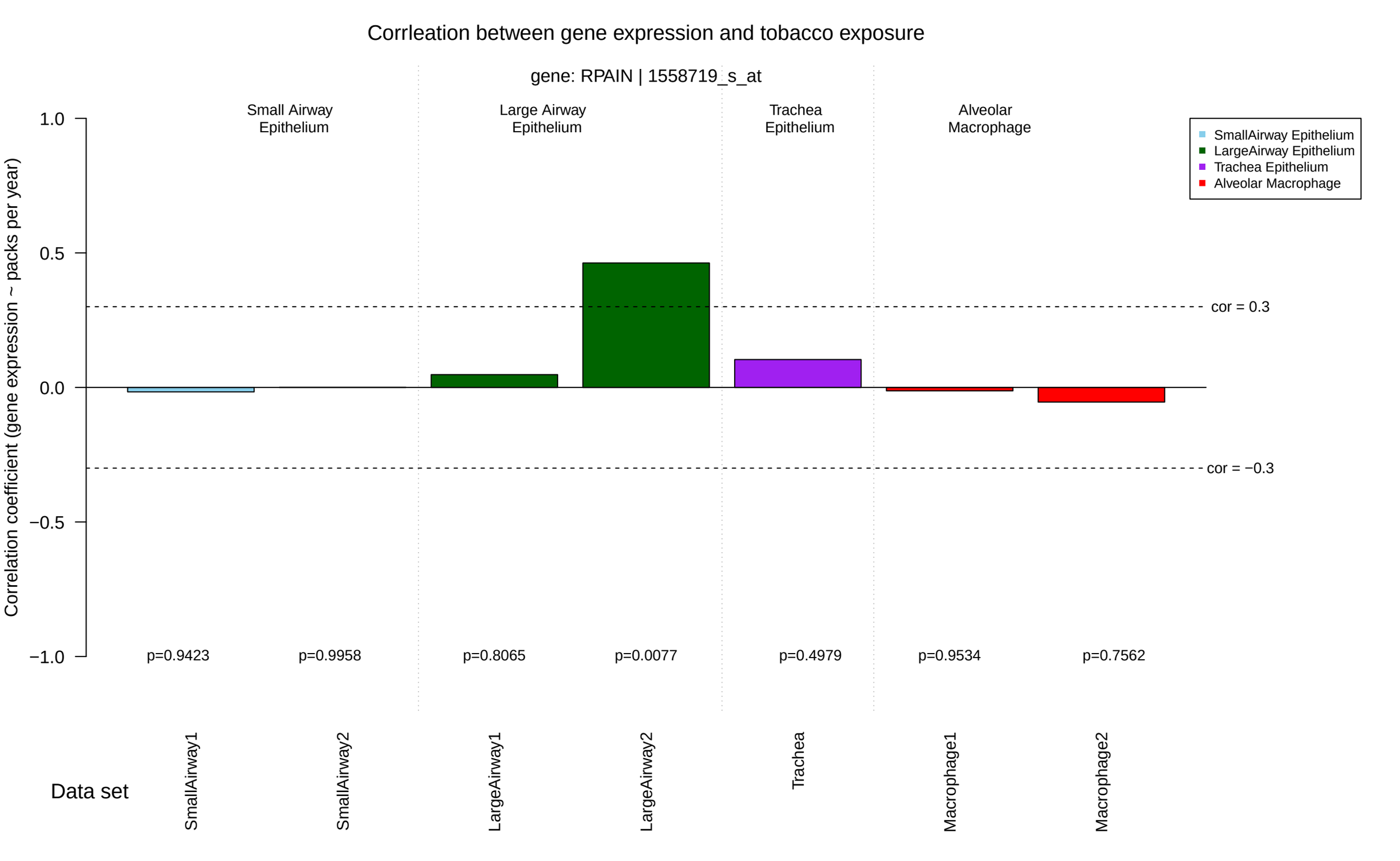

set: LargeAirway1 (p=0.7007)
RPAIN | 1558719_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.0746)
RPAIN | 1558719_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.314)
RPAIN | 1558719_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.6409)
RPAIN | 1558719_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.3013)
RPAIN | 1558719_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: RPAIN | 1558719_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.9423
p=0.9958
p=0.8065
p=0.0077
p=0.4979
p=0.9534
p=0.7562
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2